



暨南大学
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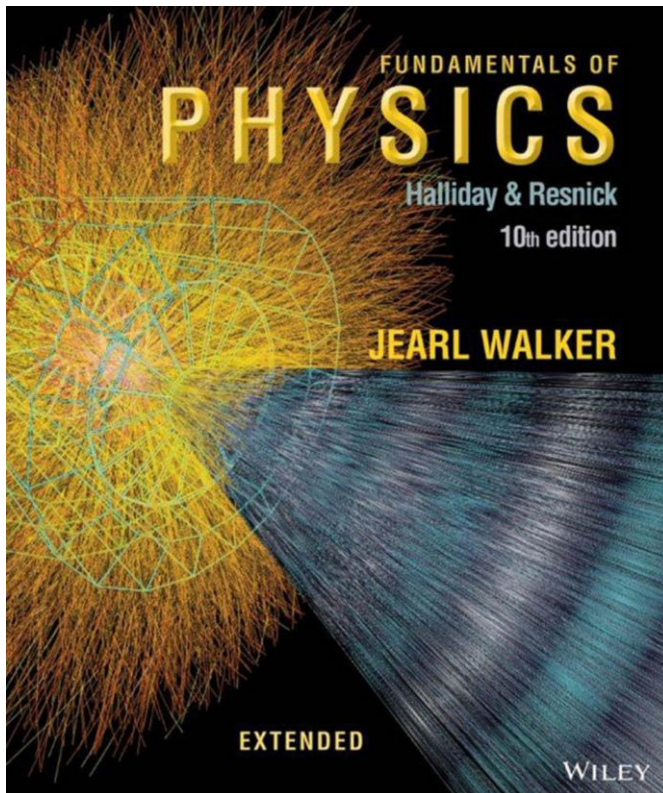
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1 class

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C H A P T E R 2 1

Coulomb's Law

- 21-1 Electric Charge
- 21-2 Conductors, Insulators and Induced Charges
- 21-3 Coulomb's Law



21-1 Electric Charge

First Observations – Greeks

- Observed electric and magnetic phenomena as early as 700 B.C.
- The Greeks found that amber, when rubbed, became electrified and attracted pieces of straw or feathers. The word “electricity” comes from the Greek word “electron”.
- Also discovered magnetic forces by observing magnetite attracting iron



Benjamin Franklin

- 1706 – 1790
- Printer, author, founding father, inventor, diplomat
- Physical Scientist
 - 1740's work on electricity changed unrelated observations into coherent science



21-1 Electric Charge

Properties of Electric Charges

- Two types of charges exist, Benjamin Franklin call them positive and negative charge, they follow:

Like charges repel, and opposite charges attract

- **Electrostatics**: The study of the interactions between charged particles **at rest**.
- **Matter** is composed of elementary particles.
- Atoms consist of **three fundamental particles**: electrons, protons, and neutrons.
- Electric charge is a property of these particles.



21-1 Electric Charge

- **Nature's basic carrier of positive charge is the proton**

Protons do not move from one material to another because they are held firmly in the nucleus

- **Nature's basic carrier of negative charge is the electron**

Gaining or losing electrons is how an object becomes charged

Two properties of Electric Charges

1. Electric charge is conserved

The algebraic sum of all the electric charge in any closed system is constant . —— Principle of conservation of charge

Electric charge can neither be created nor destroyed, but can only be transferred from one object to another, while the total amount of charge of all objects remains unchanged.

2. Electric charge is quantized

- All charge is a multiple of a fundamental unit of charge, symbolized by e
- One electron has a charge of $-e$
- One proton has a charge of $+e$
- The SI unit of charge is the Coulomb (C), $e = 1.6 \times 10^{-19} \text{ C}$



21-2 Conductors, Insulators & Induced Charges

Conductors

Conductors are materials in which the electric charges move freely in response to an electric force

- Copper, aluminum and silver are good conductors
- When a conductor is charged in a small region, the charge readily distributes itself over the entire surface of the material

Insulators

Insulators are materials in which electric charges do not move freely

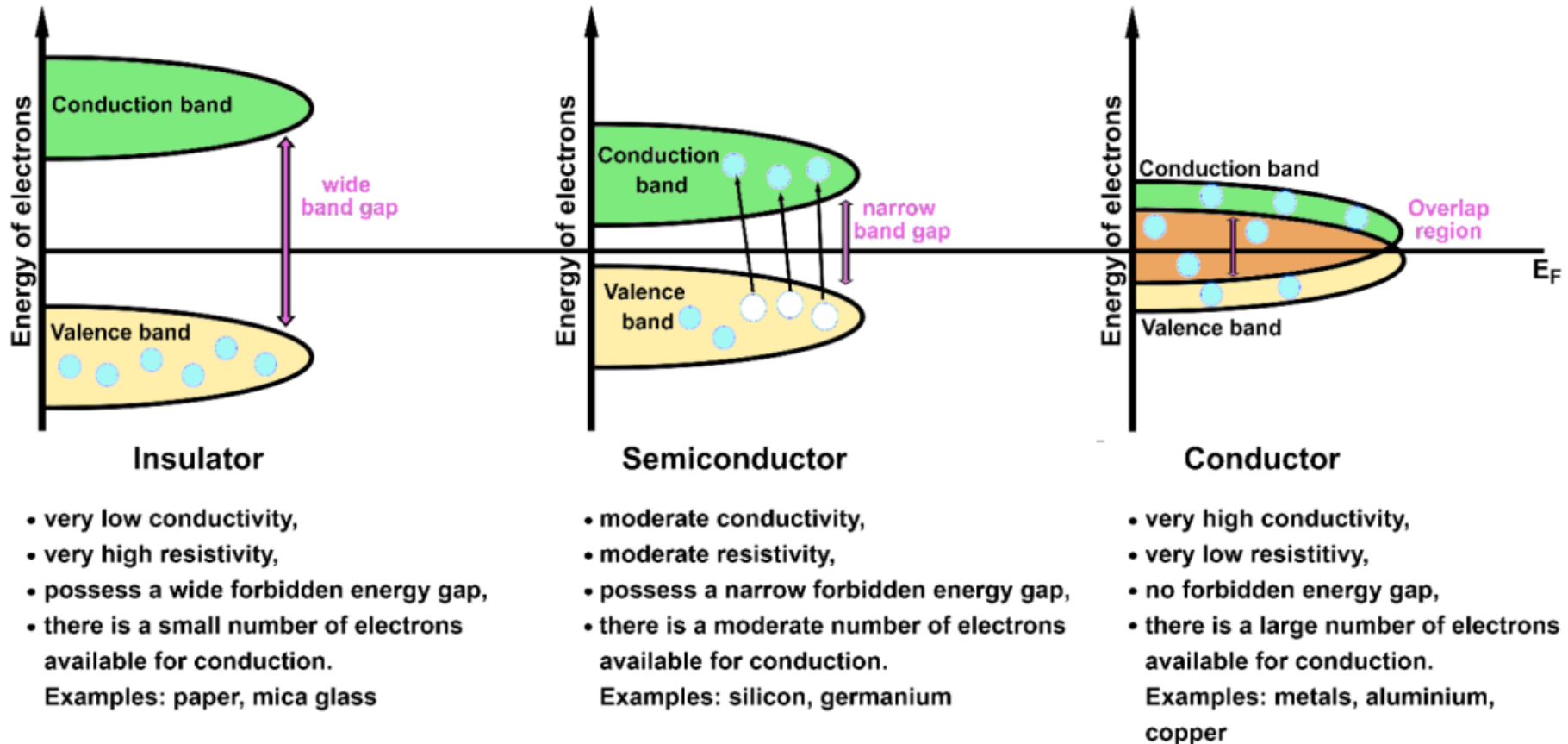
- Glass, rubber, plastic, wood are examples of insulators
- When insulators are charged by rubbing, only the rubbed area becomes charged (There is no tendency for the charge to move into other regions of the material)

Semiconductors

- The conductivity of semiconductors are between those of insulators and conductors
- Silicon (Si) and germanium (Ge) are examples of semiconductors



21-2 Conductors, Insulators & Induced Charges

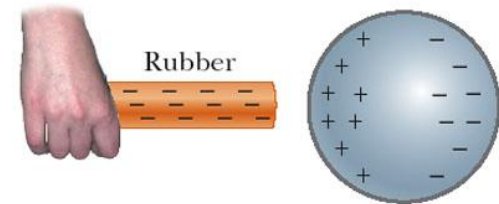


21-2 Induction & Induced Charges

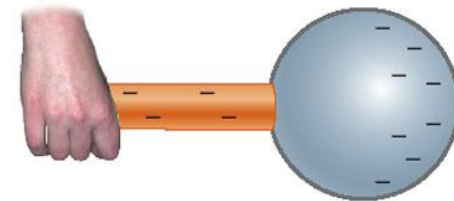
Two approaches to make a conductor becomes charged:
Conduction charging and induction charging.

1. Conduction charging

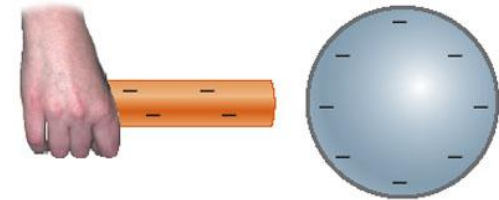
- A charged rod is placed **in contact** with the sphere
- Some electrons on the rod can move to the sphere
- When the rod is removed, the sphere is left with negative charges
- The sphere being charged is always left with a charge having the same sign as the rod doing the charging



(a) Before



(b) Contact



(c) After breaking contact

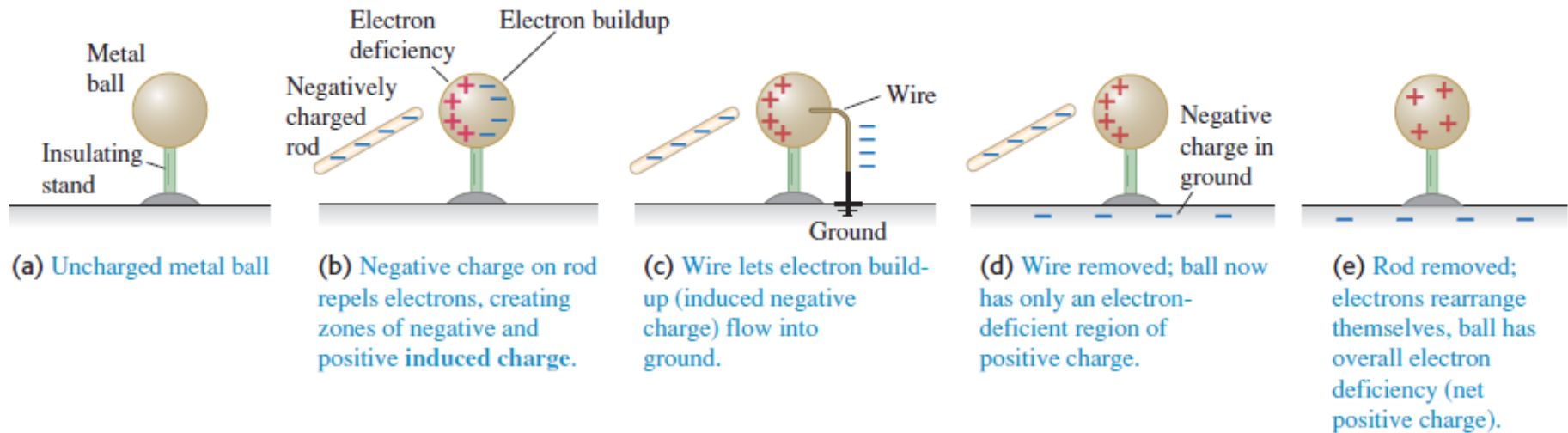


21-2 Induction & Induced Charges

Two approaches to make a conductor becomes charged:
Conduction charging and induction charging.

2. Induction charging

21.7 Charging a metal ball by induction.



21-2 Polarization effect

- In most neutral atoms or molecules, the center of positive charge coincides with the center of negative charge
- In the presence of a charged object, these centers may separate slightly
 - This results in more positive charge on one side of the molecule than on the other side
- This realignment of charge on the surface of an insulator is known as *polarization*



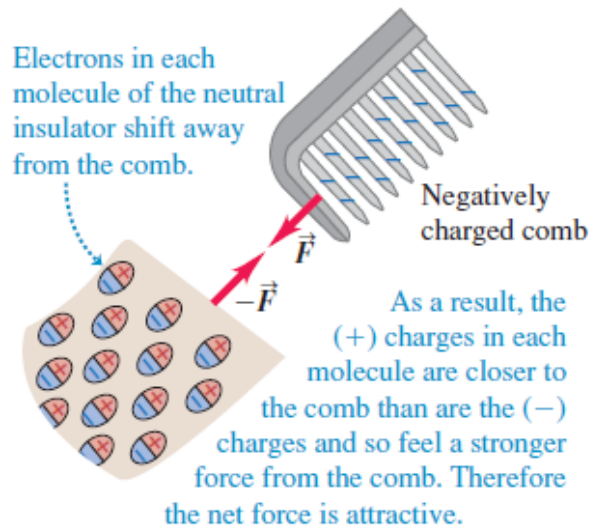
21-2 Polarization effect

21.8 The charges within the molecules of an insulating material can shift slightly. As a result, a comb with either sign of charge attracts a neutral insulator. By Newton's third law the neutral insulator exerts an equal-magnitude attractive force on the comb.

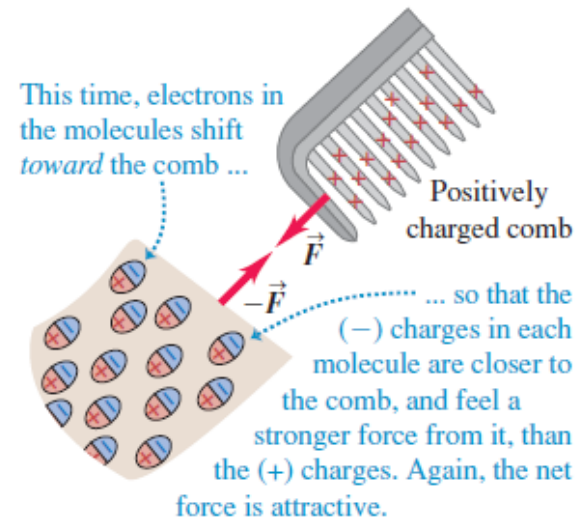
(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator



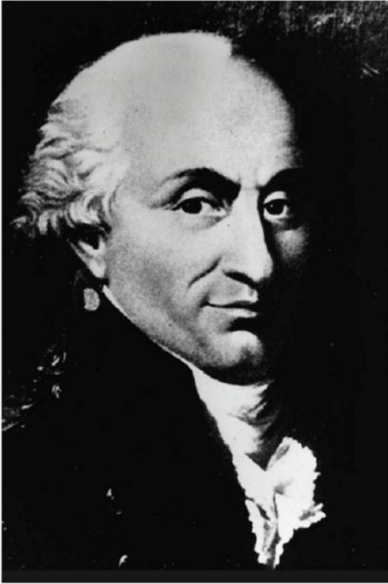
(c) How a positively charged comb attracts an insulator



- The charged object induces charge on the surface of the insulator
- A charged comb attracts bits of paper due to polarization of the paper



21-3 Coulomb's Law



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In 1784, *Charles Augustin de Coulomb* (1736-1806) used a torsion balance in the Cavendish laboratory to study the interactions between charged particles and **discovered Coulomb's Law.**



Torsion balance

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{|q_1 q_2|}{r^2}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2 ,$$

$$k = \frac{1}{4\pi\epsilon_0} \approx 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$



21-3 Coulomb's Law

- Coulomb shows that an electrical force has the following properties:
 - It is along the line joining the two particles and inversely proportional to the square of the separation distance, r , between them.
 - It is proportional to the product of the magnitudes of the point charges, $|q_1|$ and $|q_2|$ on the two particles.
 - It is attractive if the charges are of opposite signs and repulsive if the charges have the same signs.

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{|q_1 q_2|}{r^2}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2,$$

$$k = \frac{1}{4\pi\epsilon_0} \approx 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$



21-3 Coulomb's Law

- k_e is called the *Coulomb Constant*

$$k = \frac{1}{4\pi\epsilon_0} \approx 9.0 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

- Typical charges can be in the μC range
 - Remember, Coulombs must be used in the equation
- Remember that force is a **vector quantity**
- Applies only to **point charges**:
 - ✓ When the distance between charges is much larger than the size of the charged body itself, the charged body can be regarded as a point charge of negligible size;
 - ✓ The charged body itself can be regarded as composed of countless point charges.

TABLE 15.1

Charge and Mass of the Electron, Proton, and Neutron

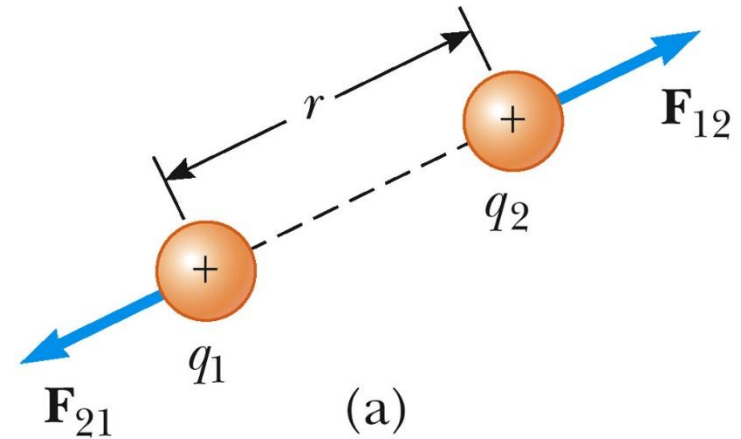
Particle	Charge (C)	Mass (kg)
Electron	-1.60×10^{-19}	9.11×10^{-31}
Proton	$+1.60 \times 10^{-19}$	1.67×10^{-27}
Neutron	0	1.67×10^{-27}



21-3 Coulomb's Law

Vector Nature of Electric Forces

- Two point charges are separated by a distance r
- The like charges produce a repulsive force between them
- The force on q_1 is equal in magnitude and opposite in direction to the force on q_2



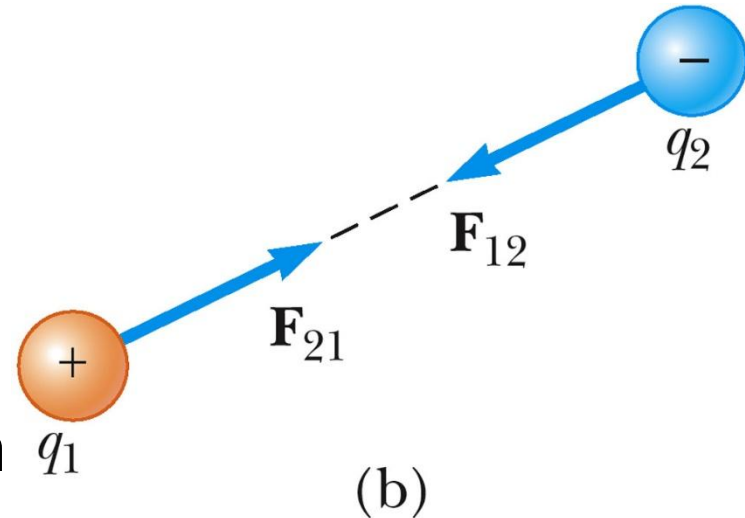
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21-3 Coulomb's Law

Vector Nature of Electric Forces

- Two point charges are separated by a distance r
- The unlike charges produce an attractive force between them
- The force on q_1 is equal in magnitude and opposite in direction to the force on q_2



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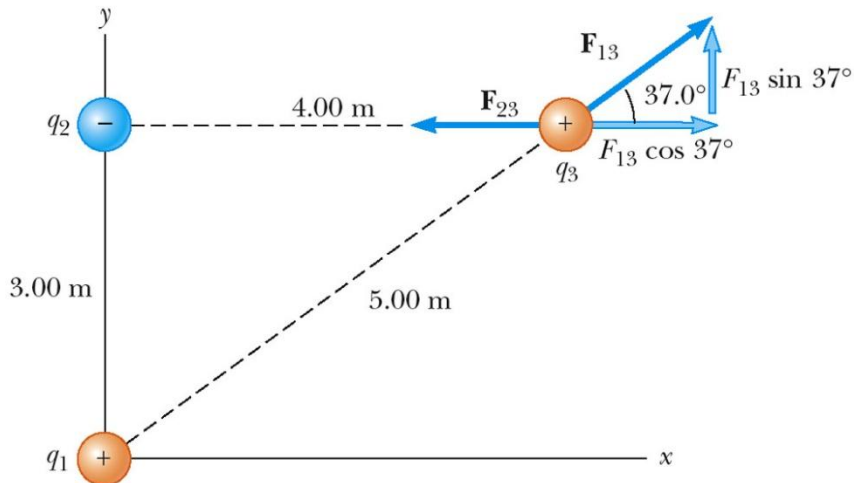


21-3 Coulomb's Law

The Superposition Principle

- The resultant force on any one charge equals the vector sum of the forces exerted by the other individual charges that are present.
 - Remember to add the forces as *vectors*

Example 1



- The force exerted by q_1 on q_3 is $\vec{F}_{13}$
- The force exerted by q_2 on q_3 is $\vec{F}_{23}$
- The *total* force exerted on q_3 is the vector sum of $\vec{F}_{13}$ and $\vec{F}_{23}$



CHAPTER 22

Electric Fields

- 22-1 Electric Field and Electric Forces
- 22-2 Electric-Field Calculations
- 22-3 Electric field lines



22-1 Electric Field and Electric Forces

The interaction between charges is **action-at-a-distance**. The transmission of the action does not pass through other substances. Instead, the **charge generates an electric field in the space around itself**. The **basic property** of the electric field is to exert a force on other charges in it.

Electric field: A physical quantity that describes the characteristics of the electric field force

$$\vec{E} = \vec{F} / q_0 \quad \left(\vec{E} = \lim_{q_0 \rightarrow 0} \vec{F} / q_0 \right) \quad [\text{Unit: N/C}]$$

The **electric field strength** is equal to the **electric force on a unit positive charge**.

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{|q_1 q_2|}{r^2}$$



22-1 Electric Field and Electric Forces

- SI unit of electric field is N / C. Use this for the magnitude of the field
- The electric field is a *vector field*
- The direction of the field is defined to be the direction of the electric force that would be exerted on a *small positive* test charge placed at that point
- An electric field with equal electric field strength everywhere is called a *uniform field*.
- Use E to calculate the electric force using $\vec{F} = q\vec{E}$
(The electric field E represents the electric force on a unit positive charge.)

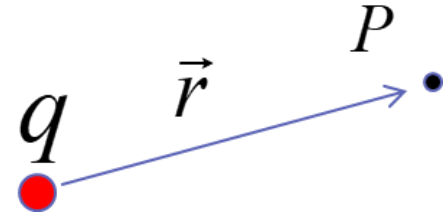


22-2 Electric Field Calculations

1. Electric field of point charge

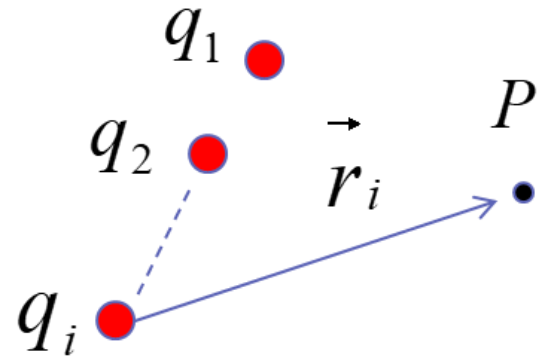
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{r}$$

The unit vector $\hat{r}$ is
 $\hat{r} = \frac{\vec{r}}{r}$



2. Electric field of discrete charges (Principle of superposition)

$$\vec{E} = \sum \vec{E}_i = \sum \frac{1}{4\pi\epsilon_0} \cdot \frac{q_i}{r_i^2} \hat{r}_i$$



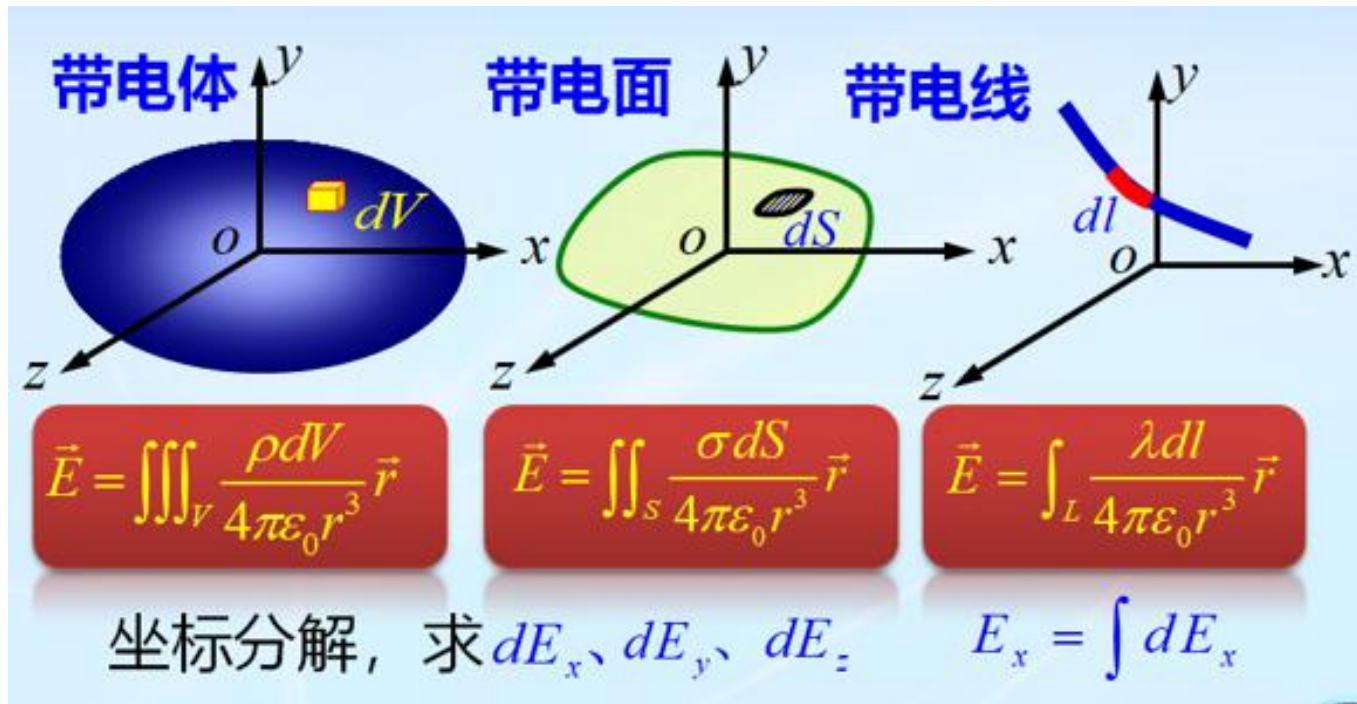
$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{|q_1 q_2|}{r^2}$$



22-2 Electric Field Calculations

3. Electric field of continuous charge distribution

Integration:
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$



- ✓ The charged body itself can be regarded as composed of countless point charges.



22-2 Electric Field Calculations

- **Problem Solving Strategy**

- **Apply Coulomb's Law**

- For each charge, find the force on the charge of interest
- Determine the direction of the force

- **Sum all the x- and y- components**

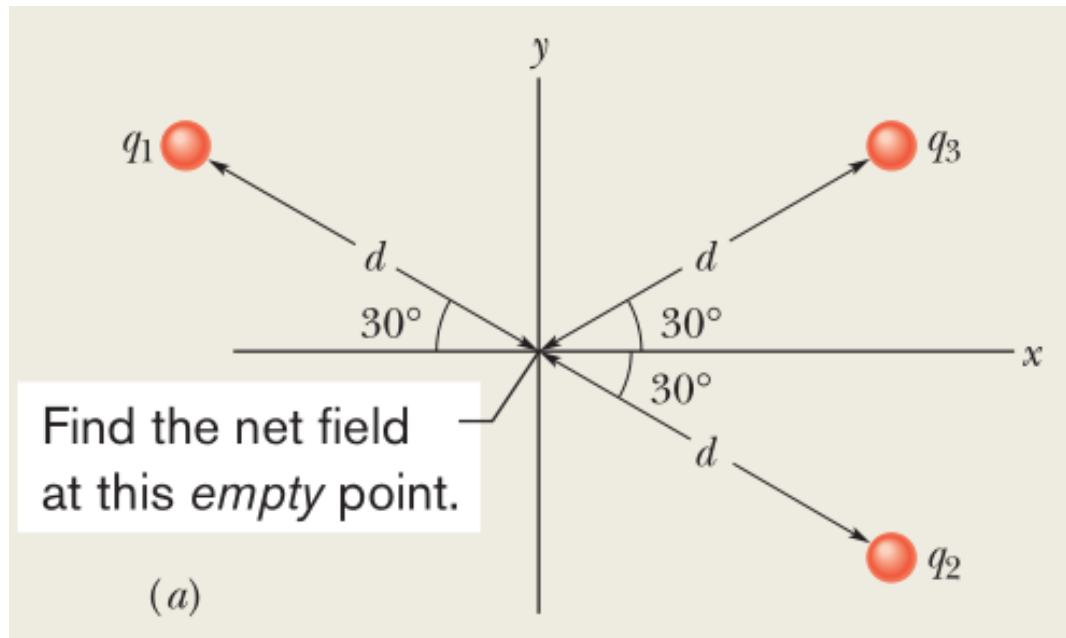
(**Convert the vector to scalars**)

- This gives the x- and y-components of the resultant force
- **Find the resultant force** by using the Pythagorean theorem and trig



Example-2 Electric field due to charged particles

Figure shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field is produced at the origin?



Example-2 Electric field due to charged particles

Fig. 1 shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field is produced at the origin?

Solution:

KEY IDEA

Charges q_1 , q_2 , and q_3 produce electric field vectors $\vec{E}_1$, $\vec{E}_2$, and $\vec{E}_3$, respectively, at the origin, and the net electric field is the vector sum $\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$. To find this sum, we first must find the magnitudes and orientations of the three field vectors.

Magnitudes and directions: To find the magnitude of $\vec{E}_1$, which is due to q_1 , we use Eq. 22-3, substituting d for r and $2Q$ for q and obtaining

$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2}.$$

Similarly, we find the magnitudes of $\vec{E}_2$ and $\vec{E}_3$ to be

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \quad \text{and} \quad E_3 = \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}.$$

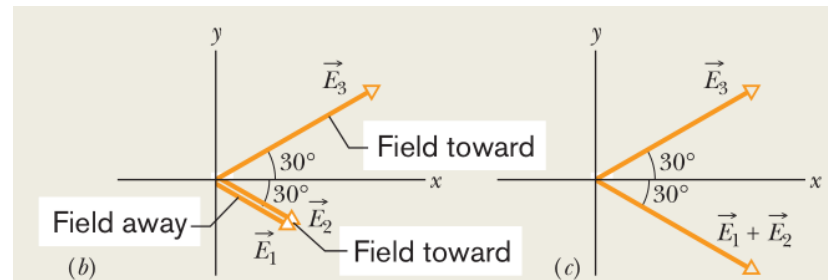


Figure 22-7 (a) Three particles with charges q_1 , q_2 , and q_3 are at the same distance d from the origin. (b) The electric field vectors $\vec{E}_1$, $\vec{E}_2$, and $\vec{E}_3$, at the origin due to the three particles. (c) The electric field vector $\vec{E}_3$ and the vector sum $\vec{E}_1 + \vec{E}_2$ at the origin.



Example-2 Electric field due to charged particles

Fig. 1 shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, and $q_3 = -4Q$, each a distance d from the origin. What net electric field is produced at the origin?

We next must find the orientations of the three electric field vectors at the origin. Because q_1 is a positive charge, the field vector it produces points directly *away* from it, and because q_2 and q_3 are both negative, the field vectors they produce point directly *toward* each of them. Thus, the three electric fields produced at the origin by the three charged particles are oriented as in Fig. 22-7b. (*Caution:* Note that we have placed the tails of the vectors at the point where the fields are to be evaluated; doing so decreases the chance of error. Error becomes very probable if the tails of the field vectors are placed on the particles creating the fields.)

Adding the fields: We can now add the fields vectorially just as we added force vectors in Chapter 21. However, here we can use symmetry to simplify the procedure. From Fig. 22-7b, we see that electric fields $\vec{E}_1$ and $\vec{E}_2$ have the same direction. Hence, their vector sum has that direction and has the magnitude

$$\begin{aligned} E_1 + E_2 &= \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} + \frac{1}{4\pi\epsilon_0} \frac{2Q}{d^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2}, \end{aligned}$$

which happens to equal the magnitude of field $\vec{E}_3$.

We must now combine two vectors, $\vec{E}_3$ and the vector sum $\vec{E}_1 + \vec{E}_2$, that have the same magnitude and that are oriented symmetrically about the x axis, as shown in Fig. 22-7c. From the symmetry of Fig. 22-7c, we realize that the equal y components of our two vectors cancel (one is upward and the other is downward) and the equal x components add (both are rightward). Thus, the net electric field $\vec{E}$ at the origin is in the positive direction of the x axis and has the magnitude

$$\begin{aligned} E &= 2E_{3x} = 2E_3 \cos 30^\circ \\ &= (2) \frac{1}{4\pi\epsilon_0} \frac{4Q}{d^2} (0.866) = \frac{6.93Q}{4\pi\epsilon_0 d^2}. \end{aligned} \quad (\text{Answer})$$



22-2 Electric Field Calculations

Problem Solving Strategy (P642-643)

Problem-Solving Tactics A Field Guide for Lines of Charge

Here is a generic guide for finding the electric field $\vec{E}$ produced at a point P by a line of uniform charge, either circular or straight. The general strategy is to pick out an element dq of the charge, find $d\vec{E}$ due to that element, and integrate $d\vec{E}$ over the entire line of charge.

Step 1. If the line of charge is circular, let ds be the arc length of an element of the distribution. If the line is straight, run an x axis along it and let dx be the length of an element. Mark the element on a sketch.

Step 2. Relate the charge dq of the element to the length of the element with either $dq = \lambda ds$ or $dq = \lambda dx$. Consider dq and λ to be positive, even if the charge is actually negative. (The sign of the charge is used in the next step.)

Step 3. Express the field $d\vec{E}$ produced at P by dq with Eq. 22-3, replacing q in that equation with either λds or λdx . If the charge on the line is positive, then at P draw a vector $d\vec{E}$ that points directly away from dq . If the charge is negative, draw the vector pointing directly toward dq .

Step 4. Always look for any symmetry in the situation. If P is on an axis of symmetry of the charge distribution, resolve the field $d\vec{E}$ produced by dq into components that are perpendicular and parallel to the axis of symmetry. Then consider a second element dq' that is located symmetrically to dq about the line of symmetry. At P draw the vector $d\vec{E}'$ that this symmetrical element pro-

duces and resolve it into components. One of the components produced by dq is a *canceled component*; it is canceled by the corresponding component produced by dq' and needs no further attention. The other component produced by dq is an *adding component*; it adds to the corresponding component produced by dq' . Add the adding components of all the elements via integration.

Step 5. Here are four general types of uniform charge distributions, with strategies for the integral of step 4.

Ring, with point P on (central) axis of symmetry, as in Fig. 22-11. In the expression for dE , replace r^2 with $z^2 + R^2$, as in Eq. 22-12. Express the adding component of $d\vec{E}$ in terms of θ . That introduces $\cos \theta$, but θ is identical for all elements and thus is not a variable. Replace $\cos \theta$ as in Eq. 22-13. Integrate over s , around the circumference of the ring.

Circular arc, with point P at the center of curvature, as in Fig. 22-13. Express the adding component of $d\vec{E}$ in terms of θ . That introduces either $\sin \theta$ or $\cos \theta$. Reduce the resulting two variables s and θ to one, θ , by replacing ds with $r d\theta$. Integrate over θ from one end of the arc to the other end.

Straight line, with point P on an extension of the line, as in Fig. 22-14a. In the expression for dE , replace r with x . Integrate over x , from end to end of the line of charge.

22-2 Electric Field Calculations

Problem Solving Strategy (P642-643)

Straight line, with point P at perpendicular distance y from the line of charge, as in Fig. 22-14b. In the expression for dE , replace r with an expression involving x and y . If P is on the perpendicular bisector of the line of charge, find an expression for the adding component of $d\vec{E}$. That will introduce either $\sin \theta$ or $\cos \theta$. Reduce the resulting two variables x and θ to one, x , by replacing the trigonometric function with an expression (its definition) involving x and y . Integrate over x from end to end of the line of charge. If P is not on a line of symmetry, as in Fig. 22-14c, set up an integral to sum the components dE_x , and integrate over x to find E_x . Also set up an integral to sum the components dE_y , and integrate over x again to find E_y . Use the components E_x and E_y in the usual way to find the magnitude E and the orientation of $\vec{E}$.

Step 6. One arrangement of the integration limits gives a positive result. The reverse gives the same result with a mi-

nus sign; discard the minus sign. If the result is to be stated in terms of the total charge Q of the distribution, replace λ with Q/L , in which L is the length of the distribution.

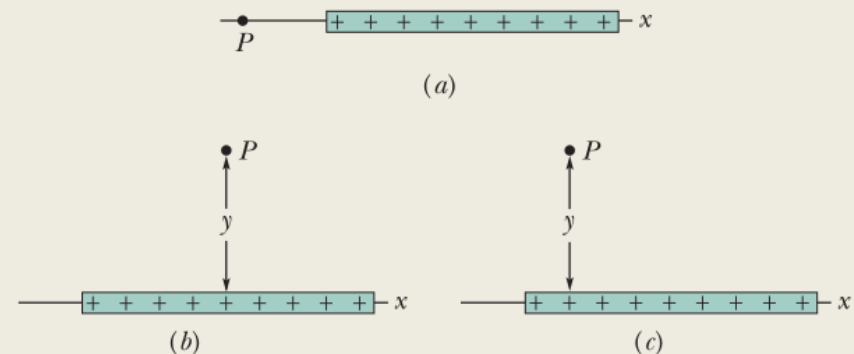


Figure 22-14 (a) Point P is on an extension of the line of charge. (b) P is on a line of symmetry of the line of charge, at perpendicular distance y from that line. (c) Same as (b) except that P is not on a line of symmetry.



Example-3 Electric field due to charged ring

2. A ring-shaped conductor with radius a carries a total positive charge Q uniformly distributed around it. Find the electric field at a point P that lies on the axis of the ring at a distance x from its center.

Solution:

$$E = E_x$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}, \quad r = \sqrt{x^2 + a^2}$$

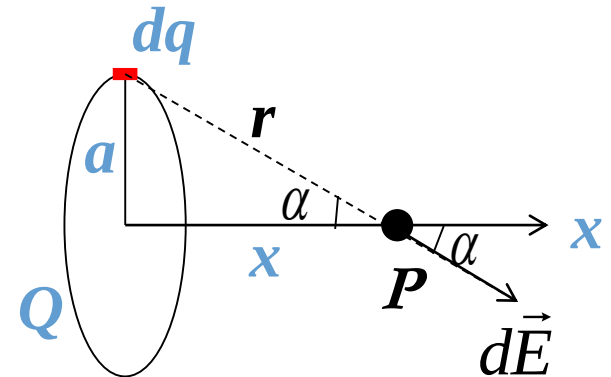
$$dE_x = dE \cdot \cos \alpha = dE \cdot \frac{x}{r} = \frac{1}{4\pi\epsilon_0} \cdot \frac{xdq}{r^3}$$

$$E_x = \oint dE_x = \frac{1}{4\pi\epsilon_0} \oint \frac{xdq}{(x^2 + a^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{xQ}{(x^2 + a^2)^{3/2}}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Qx}{(x^2 + a^2)^{3/2}} \vec{i}$$

$$x \gg a, \quad E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{x^2}$$

(Electric field of point charge)



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$



Example-4 Electric field of a charged disk

3. Find the electric field caused by a disk of radius R with a uniform positive surface charge density σ , at a point along the axis of the disk a distance x from its center. Assume that x is positive.

Solution:

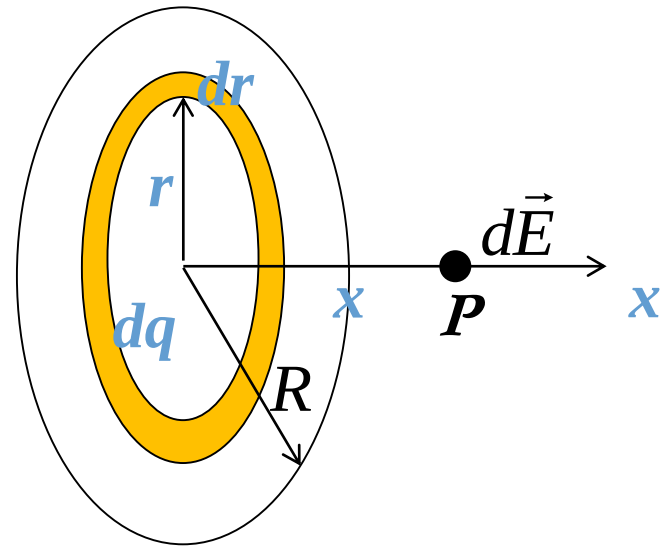
$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{x dq}{(x^2 + r^2)^{3/2}} \vec{i}, \quad dq = \sigma \cdot 2\pi r dr$$

$$E = \int dE = \int_0^R \frac{1}{4\pi\epsilon_0} \cdot \frac{x \cdot \sigma 2\pi r dr}{(x^2 + r^2)^{3/2}}$$

$$= \frac{\sigma x}{2\epsilon_0} \left(\frac{1}{x} - \frac{1}{\sqrt{x^2 + R^2}} \right) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{(R/x)^2 + 1}} \right)$$

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{1}{\sqrt{(R/x)^2 + 1}} \right) \vec{i}$$

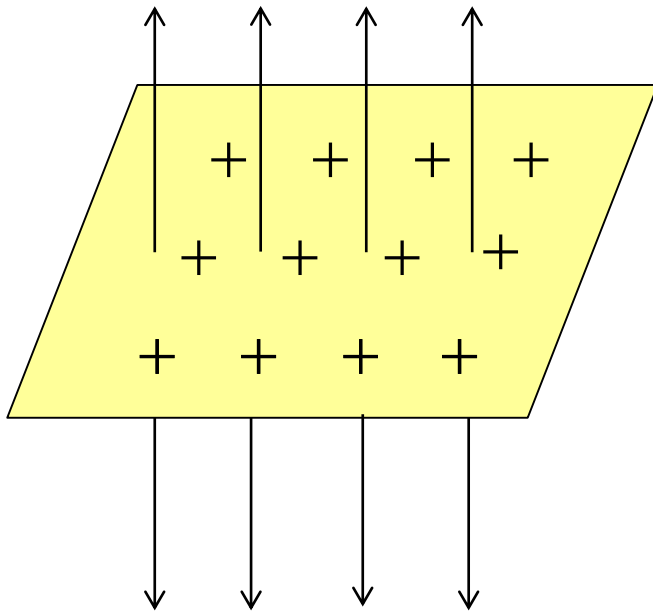
$$x \ll R, \quad \vec{E} = \frac{\sigma}{2\epsilon_0} \vec{i}$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Qx}{(x^2 + a^2)^{3/2}} \vec{i}$$



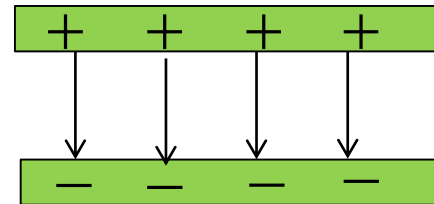
Expand



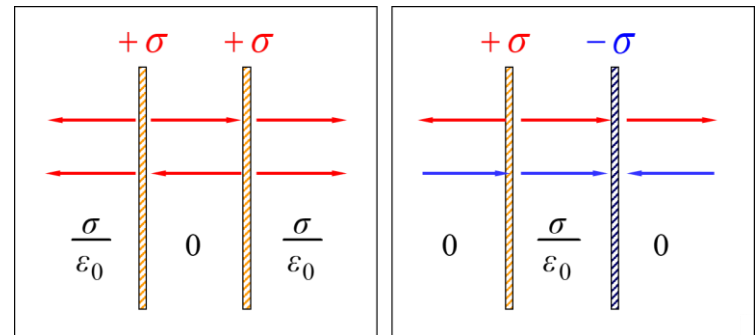
$$E = \frac{\sigma}{2\epsilon_0}$$

The electric field strength generated by a flat plate **with infinite charges** evenly distributed in space is a **uniform field**, and the **direction of the field strength** is **perpendicular to the plane**.

Parallel capacitor



$$E = \frac{\sigma}{\epsilon_0}$$



22-3 Electric Field Lines

The spatial distribution of the electric field can be vividly described using **electric field lines** (line of force/field). A series of directional curves are drawn in the electric field. The direction of the tangent line represents the direction of the electric field strength at that point, and the density of the curves represents the magnitude of the electric field strength at that point.

Characteristics of electric field lines:

1. Starting from a positive charge and returning to a negative charge, the curve is an open line.
2. The curves do not intersect in the uncharged region.

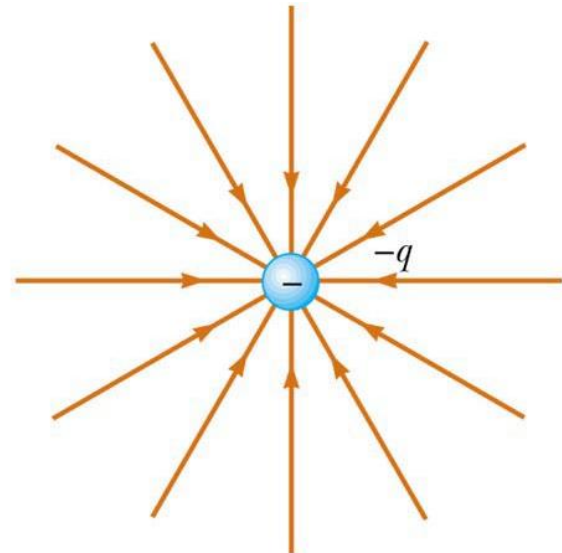
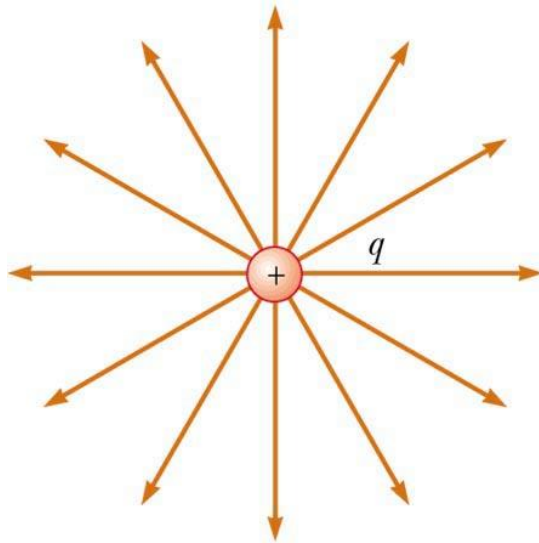
 The electric field lines of a uniform electric field are parallel lines with equal spacing.

 Charged particles do not necessarily move along electric field lines. Electric field lines are artificially introduced and do not actually exist.



22-3 Electric Field Lines

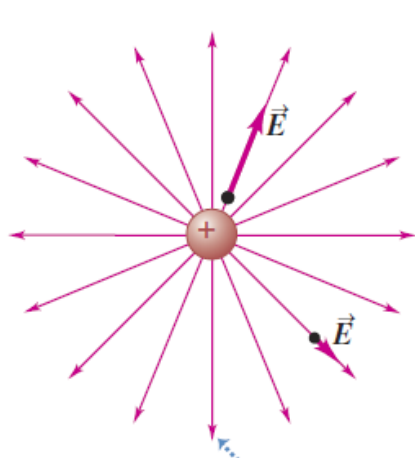
- Point charge
- The lines radiate equally in all directions
- For a **positive** source charge, the lines will radiate outward
- For a **negative** source charge, the lines will point inward



22-3 Electric Field Lines

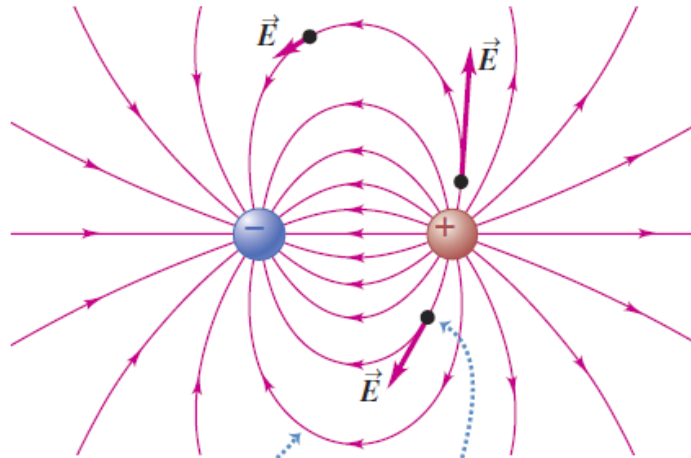
21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.

(a) A single positive charge



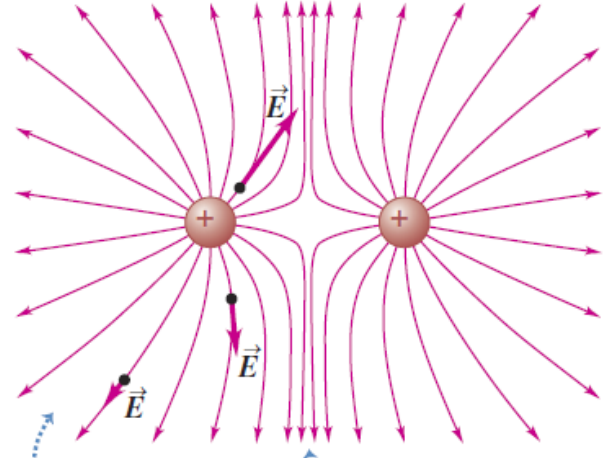
Field lines always point away from (+) charges and toward (-) charges.

(b) Two equal and opposite charges (a dipole)



At each point in space, the electric field vector is *tangent* to the field line passing through that point.

(c) Two equal positive charges



Field lines are close together where the field is strong, farther apart where it is weaker.



Gauss' Law

- 23-1 Electric flux and Enclosed Charge
- 23-2 Calculating electric flux
- 23-3 Gauss's Law
- 23-4 Applications of Gauss's Law



23-1 Electric flux and Enclosed Charge



**Carl Friedrich
Gauss
1777–1855**

In this chapter we discuss a relationship between charge and electric field that allows us, in certain symmetric situations, to find the electric field of an extended charged object with a few lines of algebra.

The relationship is called **Gauss' law**, which was developed by German mathematician and physicist Carl Friedrich Gauss.



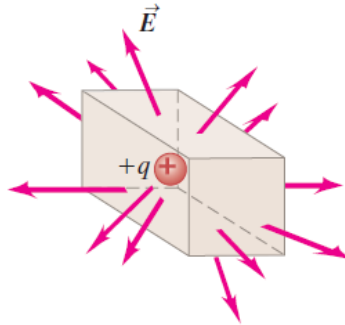
23-1 Electric flux and Enclosed Charge

Electric flux: Number of electric field lines passing through the curved surface.

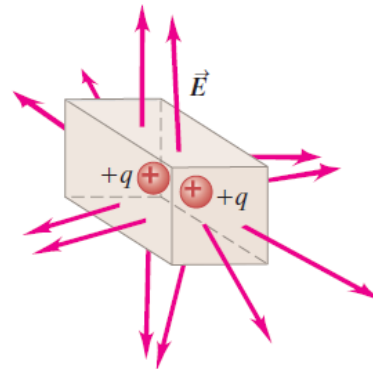
- The density of electric field lines reflects the magnitude of the electric field strength, which is the surface density of the electric field lines.

22.2 The electric field on the surface of boxes containing (a) a single positive point charge, (b) two positive point charges, (c) a single negative point charge, or (d) two negative point charges.

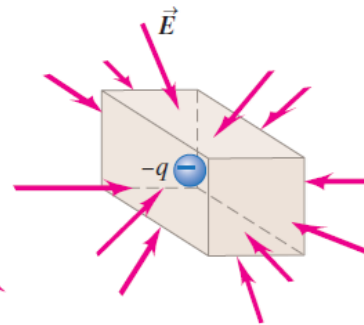
(a) Positive charge inside box, outward flux



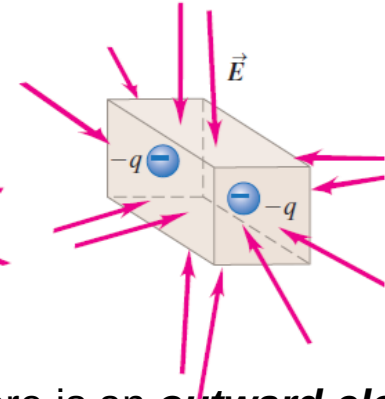
(b) Positive charges inside box, outward flux



(c) Negative charge inside box, inward flux



(d) Negative charges inside box, inward flux



- The electric field vectors point out of the surface, we say that there is an **outward electric flux**. (The word “flux” comes from a Latin word meaning “flow.”)
- In Figs. 22.2c and 22.2d, the vectors point into the surface, and the **electric flux is inward**.
- This figure suggests a simple relationship:

Positive charge inside the box goes with an outward electric flux through the box's surface, and negative charge inside goes with an inward electric flux.



23-1 Electric flux and Enclosed Charge

To summarize, for the special cases of a closed surface in the shape of a rectangular box and charge distributions made up of point charges or infinite charged sheets, we have found:

1. Whether there is a **net outward or inward electric flux** through a closed surface **depends on the sign of the enclosed charge**.
2. Charges outside the surface do not give a net electric flux through the surface.
3. The net electric flux is directly proportional to the net amount of charge enclosed within the surface but is independent of the size and shape of the closed surface.

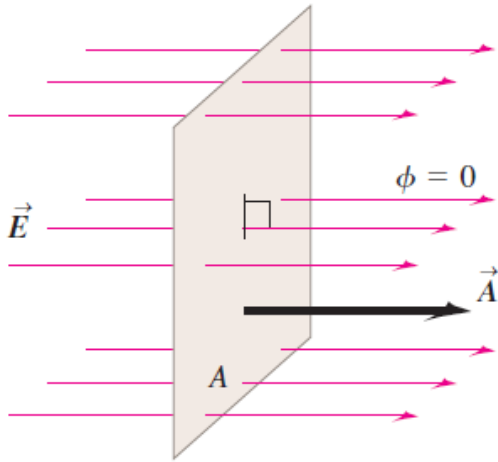


23-2 Calculating electric flux

Situation 1: Flat Surface, Uniform Field

(a) Surface is face-on to electric field:

- $\vec{E}$ and $\vec{A}$ are parallel (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 0$).
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA$.



The symbol that we use for **electric flux** is Φ_E (the capital Greek letter phi; the subscript E is a reminder that this is electric flux).

Consider first a flat area perpendicular to a uniform electric field (Figure a). We define the **electric flux through this area** to be the product of the field magnitude E and the area A :

$$\Phi_E = EA$$

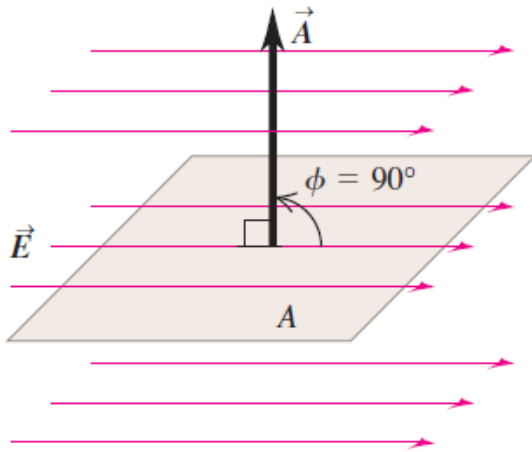
Roughly speaking, we can picture Φ_E in terms of the field lines passing through A . Increasing the area means that more lines of $\vec{E}$ pass through the area, increasing the flux; a stronger field means more closely spaced lines of $\vec{E}$ and therefore more lines per unit area, so again the flux increases.



23-2 Calculating electric flux

Situation 1: Flat Surface, Uniform Field

- (c) Surface is edge-on to electric field:
- $\vec{E}$ and $\vec{A}$ are perpendicular (the angle between $\vec{E}$ and $\vec{A}$ is $\phi = 90^\circ$).
 - The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos 90^\circ = 0$.



If the area is edge-on to the field, $\vec{E}$ and $\vec{A}$ are perpendicular and the flux is zero.

We can represent the direction of a vector area $\vec{A}$ by using a *unit vector* $\hat{n}$ perpendicular to the area; $\hat{n}$ stands for “normal.” Then

$$\vec{A} = A\hat{n}$$

A surface has two sides, so there are two possible directions for $\hat{n}$ and $\vec{A}$. We must always specify which direction we choose. We have related the charge inside a *closed surface* to the electric flux through the surface. **With a closed surface we will always choose the direction of $\hat{n}$ to be outward,** and we will speak of the **flux out of a closed surface.**

Thus what we called “**outward electric flux**” corresponds to a **positive value** of Φ_E , and what we called “**inward electric flux**” corresponds to a **negative value** of Φ_E .

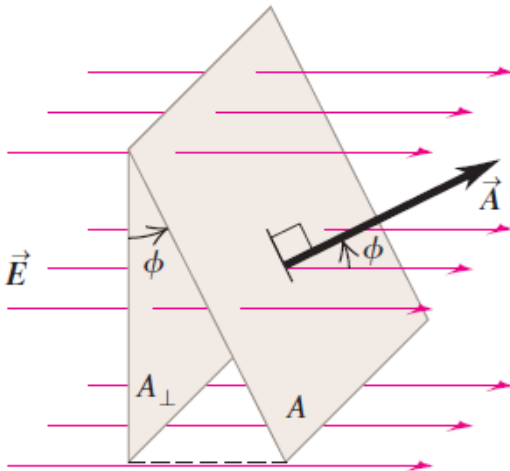


23-2 Calculating electric flux

Situation 1: Flat Surface, Uniform Field

(b) Surface is tilted from a face-on orientation by an angle ϕ :

- The angle between $\vec{E}$ and $\vec{A}$ is ϕ .
- The flux $\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \phi$.



If the area A is flat but not perpendicular to the field $\vec{E}$, then fewer field lines pass through it.

In this case the area that counts is the profile area (projection surface) that we see when looking in the direction of $\vec{E}$. This is the area $A_{\perp}$ in Fig. b and is equal to $A \cos \Phi$. We generalize our definition of electric flux for a uniform electric field to:

$$\Phi_E = EA \cos \phi \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$

Since $E \cos \Phi$ is the component of $\vec{E}$ perpendicular to the area, we can rewrite as:

$$\Phi_E = E_{\perp} A \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$

In terms of the vector area $\vec{A}$ perpendicular to the area, we can write the electric flux as the scalar product of $\vec{E}$ and $\vec{A}$.

$$\Phi_E = \vec{E} \cdot \vec{A} \quad (\text{electric flux for uniform } \vec{E}, \text{ flat surface})$$



23-2 Calculating electric flux

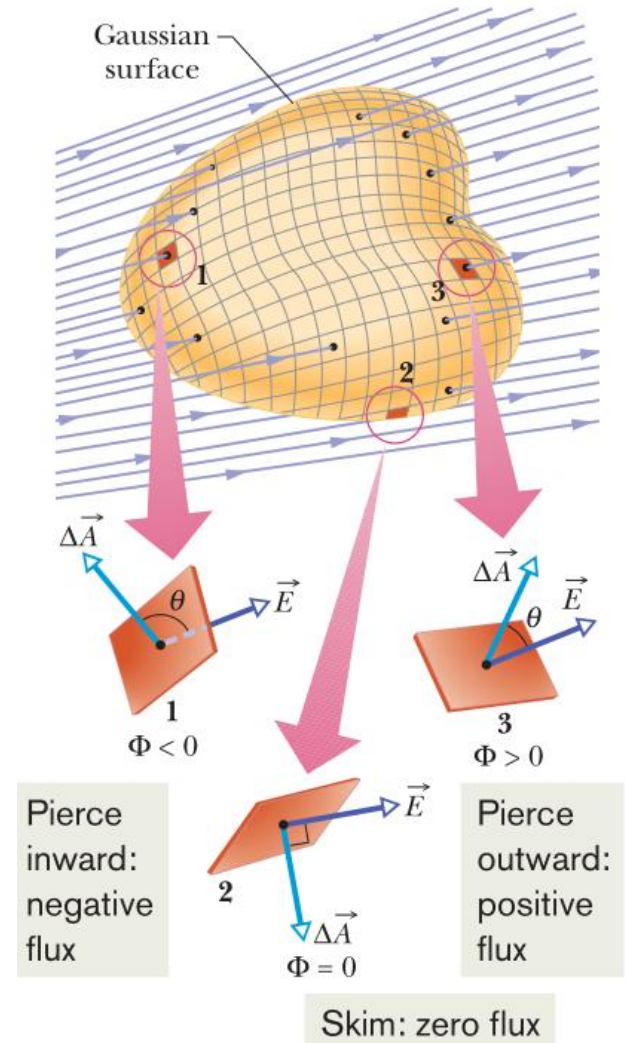
Situation 2: Flux of a Nonuniform Electric Field

What happens if the electric field $\vec{E}$ isn't uniform but varies from point to point over the area A ? Or what if A is part of a curved surface?

We divide A into many small elements dA , each of which has a unit vector $\hat{n}$ perpendicular to it and a vector area $d\vec{A} = \hat{n} dA$. We calculate the electric flux through each element and integrate the results to obtain the total net flux:

$$\Phi = \oint \vec{E} \cdot d\vec{A} \quad (\text{net flux}).$$

We call this integral the **surface integral** of the component $E_{\perp}$ over the area, or the surface integral of $\vec{E} \cdot d\vec{A}$.



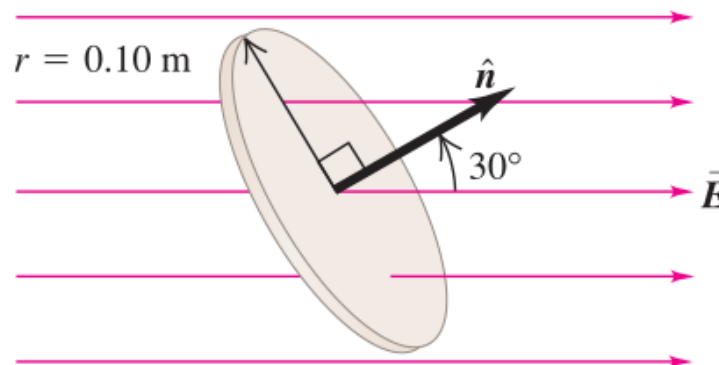
Example-5

Electric flux through a disk

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude 2.0×10^3 N/C as shown in the Figure. (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in this open surface.)

- (a) What is the electric flux through the disk?
- (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$?
- (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?

22.7 The electric flux Φ_E through a disk depends on the angle between its normal $\hat{n}$ and the electric field $\vec{E}$.



Example-5

SOLUTION

IDENTIFY and SET UP: This problem is about a flat surface in a uniform electric field, so we can apply the ideas of this section. We calculate the electric flux using Eq. (22.1).

EXECUTE: (a) The area is $A = \pi(0.10 \text{ m})^2 = 0.0314 \text{ m}^2$ and the angle between $\vec{E}$ and $\vec{A} = A\hat{n}$ is $\phi = 30^\circ$, so from Eq. (22.1),

$$\begin{aligned}\Phi_E &= EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(\cos 30^\circ) \\ &= 54 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

(b) The normal to the disk is now perpendicular to $\vec{E}$, so $\phi = 90^\circ$, $\cos \phi = 0$, and $\Phi_E = 0$.

(c) The normal to the disk is parallel to $\vec{E}$, so $\phi = 0$ and $\cos \phi = 1$:

$$\begin{aligned}\Phi_E &= EA \cos \phi = (2.0 \times 10^3 \text{ N/C})(0.0314 \text{ m}^2)(1) \\ &= 63 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

EVALUATE: As a check on our results, note that our answer to part (b) is smaller than that to part (a), which is in turn smaller than that to part (c). Is all this as it should be?



Example-6

Electric flux through a sphere

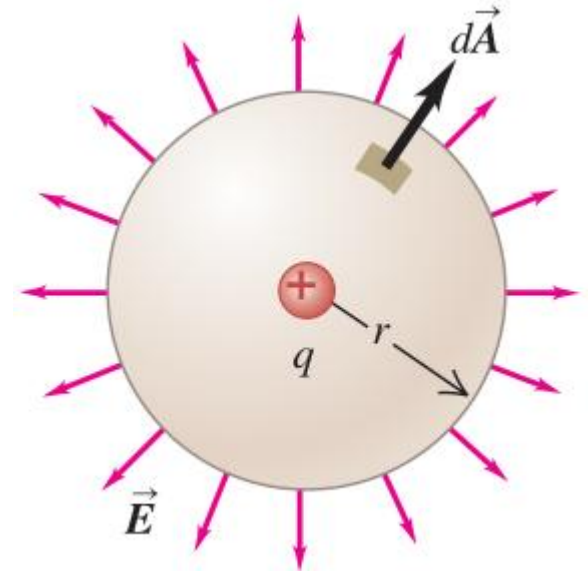
A point charge $q = +3.0 \mu\text{C}$ is surrounded by an imaginary sphere of radius $r = 0.20 \text{ m}$ centered on the charge as shown in the figure. Find the resulting electric flux through the sphere.

Solution

The surface is not flat and the electric field is not uniform, so to calculate the electric flux. We must use the general definition.

Because the sphere is centered on the point charge, at any point on the spherical surface, $\vec{E}$ is directed out of the sphere perpendicular to the surface.

The positive direction for both $\hat{n}$ and $E_{\perp}$ is outward, so $E_{\perp} = E$ and the flux through a surface element dA is $\vec{E} \cdot d\vec{A} = E dA$.



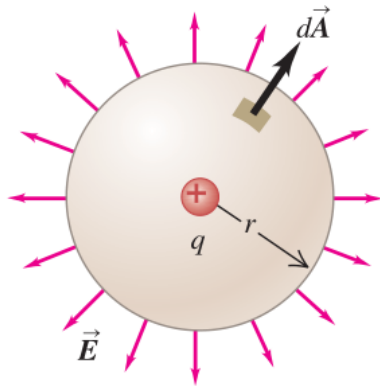
$$\Phi = \oint \vec{E} \cdot d\vec{A} \quad (\text{net flux}).$$



Example-6

Electric flux through a sphere

22.9 Electric flux through a sphere centered on a point charge.



EXECUTE: We must evaluate the integral of Eq. (22.5), $\Phi_E = \int E \, dA$. At any point on the sphere of radius r the electric field has the same magnitude $E = q/4\pi\epsilon_0 r^2$. Hence E can be taken outside the integral, which becomes $\Phi_E = E \int dA = EA$, where A is the

area of the spherical surface: $A = 4\pi r^2$. Hence the total flux through the sphere is

$$\begin{aligned}\Phi_E &= EA = \frac{q}{4\pi\epsilon_0 r^2} 4\pi r^2 = \frac{q}{\epsilon_0} \\ &= \frac{3.0 \times 10^{-6} \text{ C}}{8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2} = 3.4 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}\end{aligned}$$

EVALUATE: The radius r of the sphere cancels out of the result for Φ_E . We would have obtained the same flux with a sphere of radius 2.0 m or 200 m. We came to essentially the same conclusion in our discussion of Fig. 22.4 in Section 22.1, where we considered rectangular closed surfaces of two different sizes enclosing a point charge. There we found that the flux of $\vec{E}$ was independent of the size of the surface; the same result holds true for a spherical surface. Indeed, the flux through *any* surface enclosing a single point charge is independent of the shape or size of the surface, as we'll soon see.



23-3 Gauss's Law

Gauss's law is an alternative to Coulomb's law. While completely equivalent to Coulomb's law, Gauss's law provides a different way to express the relationship between electric charge and electric field.

Gauss' law relates the net flux Φ of an electric field through a closed surface (a Gaussian surface) to the *net* charge q_{enc} that is *enclosed* by that surface.

Gauss' law can also be written in terms of the electric field piercing the enclosing Gaussian surface:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0} \quad (\text{Gauss's Law})$$

The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0 .



23-3 Gauss's Law

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} \quad (\text{Gauss's Law})$$

The total electric flux through a closed surface is equal to the total (net) electric charge inside the surface, divided by ϵ_0 . (Gauss's Law)

where ϵ_0 is the vacuum permittivity of free space and equals $8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

- ✓ In **Gauss's Law**, the electric field **E** is the total electric field on the surface, including the electric field generated by both inside and outside surface charges.
- ✓ Q_{enc} is the algebraic sum of all charges within the surface.
- ✓ The surface S in the formula is called the *Gaussian surface*.



23-3 Gauss's Law

• Problem Solving Strategy using Gauss's Law

Step 1

Analyze the symmetry of charge distribution in a charged body, then analyze the symmetry of the electric field (i.e.: spherical symmetry (select sphere as Gaussian surface), axial symmetry (cylindrical surface), surface symmetry, etc.)

Step 2

Establish a suitable coordinate system (Polar coordinates or rectangular coordinate system)

Step 3

Select a suitable Gaussian surface (the point of interest lies on the Gaussian surface)

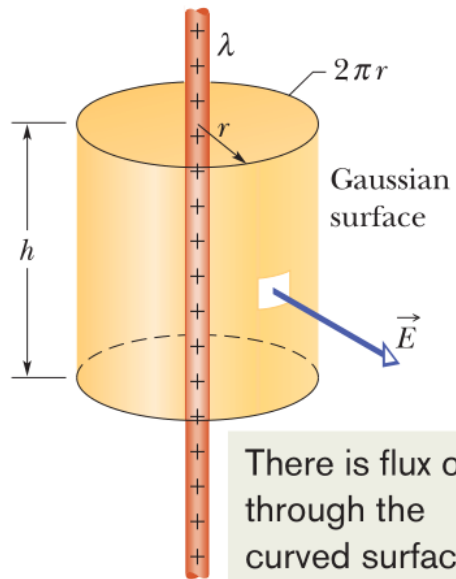
Step 4

Apply Gauss's Law to calculate the electric field strength.



23-4 Applying Gauss's Law

1. Cylindrical Symmetry



1.1 Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ . Find the expression for the electric field magnitude E at radius r from the central axis of the rod, outside the rod.

Solution:

The charge distribution and the field have cylindrical symmetry. To find the field at radius r , we enclose a section of the rod with a concentric Gaussian cylinder of radius r and height h . (If you want the field at a certain point, put a Gaussian surface through that point.) We can now apply Gauss' law to relate the charge enclosed by the cylinder and the net flux through the cylinder's surface.

First note that because of the symmetry, the electric field at any point must be radially outward (the charge is positive). That means that at any point on the end caps, the field only skims the surface and does not pierce it. So, the flux through each end cap is zero.



23-4 Applying Gauss's Law

To find the flux through the cylinder's curved surface, first note that for any patch element on the surface, the area vector $d\vec{A}$ is radially outward (away from the interior of the Gaussian surface) and thus in the same direction as the field piercing the patch. The dot product in Gauss' law is then simply $E dA \cos 0 = E dA$, and we can pull E out of the integral. The remaining integral is just the instruction to sum the areas of all patch elements on the cylinder's curved surface, but we already know that the total area is the product of the cylinder's height h and circumference $2\pi r$. The net flux through the cylinder is then

$$\Phi = EA \cos \theta = E(2\pi rh) \cos 0 = E(2\pi rh).$$

On the other side of Gauss's law we have the charge q_{enc} enclosed by the cylinder. Because the linear charge density (charge per unit length, remember) is uniform, the enclosed charge is λh . Thus, Gauss' law,

$$\epsilon_0 \Phi = q_{\text{enc}},$$

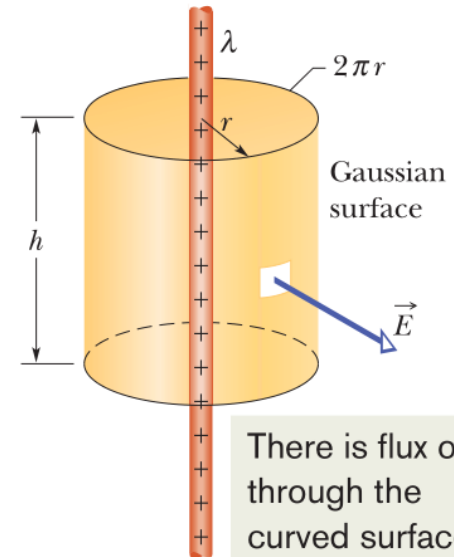
reduces to

$$\epsilon_0 E(2\pi rh) = \lambda h,$$

yielding

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (\text{line of charge}). \quad (23-12)$$

This is the electric field due to an infinitely long, straight line of charge, at a point that is a radial distance r from the line. The direction of $\vec{E}$ is radially outward from the line of charge if the charge is positive, and radially inward if it is negative. Equation 23-12 also approximates the field of a *finite* line of charge at points that are not too near the ends (compared with the distance from the line).

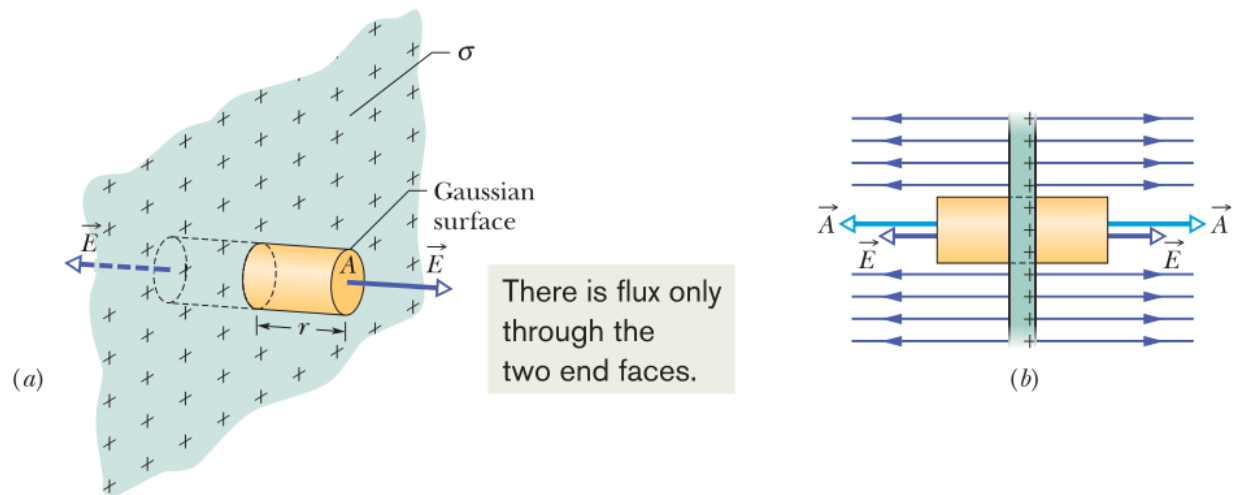


23-4 Applying Gauss's Law

2. Planar Symmetry

2.1 Figure 23-17 shows a portion of a thin, infinite, **nonconducting sheet** with a uniform positive surface charge density σ . A sheet of thin plastic wrap, uniformly charged on one side, can serve as a simple model. Find the electric field $\vec{E}$ a distance r in front of the sheet.

Figure 23-17 (a) Perspective view and (b) side view of a portion of a very large, thin plastic sheet, uniformly charged on one side to surface charge density σ . A closed cylindrical Gaussian surface passes through the sheet and is perpendicular to it.



23-4 Applying Gauss's Law

2. Planar Symmetry

A useful Gaussian surface is a closed cylinder with end caps of area A , arranged to pierce the sheet perpendicularly as shown. From symmetry, $\vec{E}$ must be perpendicular to the sheet and hence to the end caps. Furthermore, since the charge is positive, $\vec{E}$ is directed *away* from the sheet, and thus the electric field lines pierce the two Gaussian end caps in an outward direction. Because the field lines do not pierce the curved surface, there is no flux through this portion of the Gaussian surface. Thus $\vec{E} \cdot d\vec{A}$ is simply $E dA$; then Gauss' law,

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q_{\text{enc}},$$

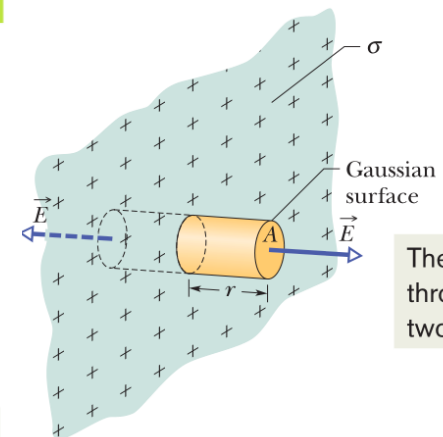
becomes

$$\epsilon_0(EA + EA) = \sigma A,$$

where σA is the charge enclosed by the Gaussian surface. This gives

$$E = \frac{\sigma}{2\epsilon_0} \quad (\text{sheet of charge}). \quad (23-13)$$

Since we are considering an infinite sheet with uniform charge density, this result holds for any point at a finite distance from the sheet. Equation 23-13 agrees with Eq. 22-27, which we found by integration of electric field components.



23-4 Applying Gauss's Law

2. Planar Symmetry

Suppose we arrange for the plates of Figs. 23-18*a* and *b* to be close to each other and parallel (Fig. 23-18*c*). Since the plates are conductors, when we bring them into this arrangement, the excess charge on one plate attracts the excess charge on the other plate, and all the excess charge moves onto the inner faces of the plates as in Fig. 23-18*c*. With twice as much charge now on each inner face, the new surface charge density (call it σ) on each inner face is twice σ_1 . Thus, the electric field at any point between the plates has the magnitude

$$E = \frac{2\sigma_1}{\epsilon_0} = \frac{\sigma}{\epsilon_0}. \quad (23-14)$$

This field is directed away from the positively charged plate and toward the negatively charged plate. Since no excess charge is left on the outer faces, the electric field to the left and right of the plates is zero.

$$E = \frac{2\sigma_1}{\epsilon_0} = \frac{\sigma}{\epsilon_0}.$$

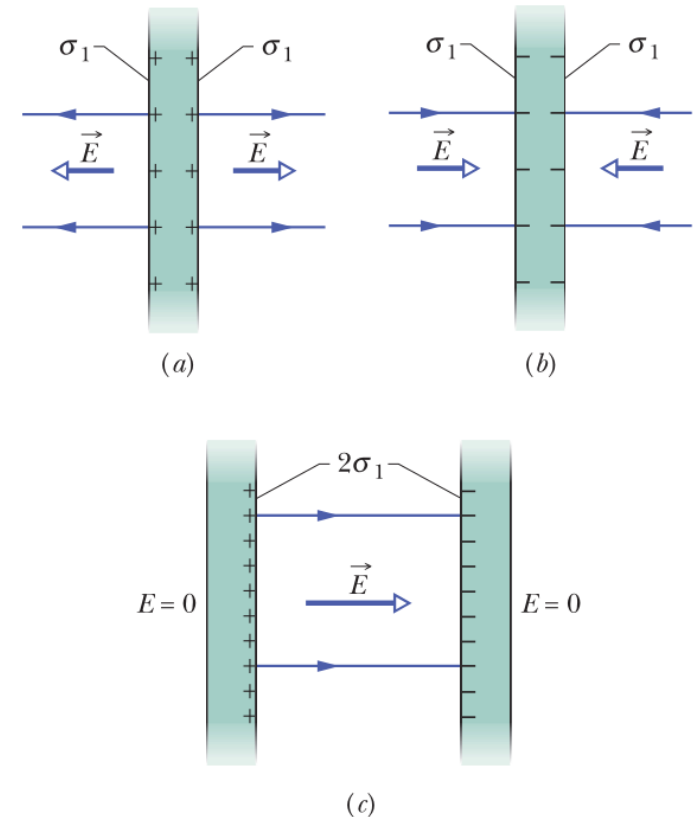


Figure 23-18 (a) A thin, very large conducting plate with excess positive charge. (b) An identical plate with excess negative charge. (c) The two plates arranged so they are parallel and close.



23-4 Applying Gauss's Law

3. Spherical Symmetry

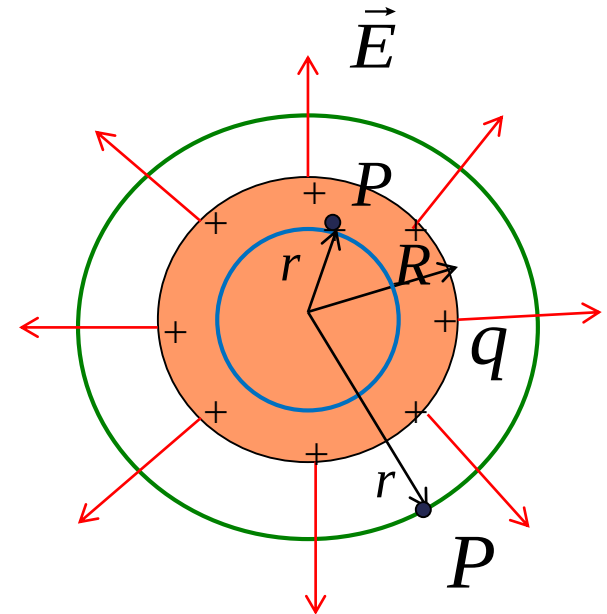
3.1 We place positive charge q on a **solid conducting sphere** with radius R . Find E at any point inside or outside the sphere.

Solution: $r < R, \oint \vec{E} \cdot d\vec{A} = 0, \vec{E} = 0$

$$r > R, \oint \vec{E} \cdot d\vec{A} = q / \epsilon_0$$

$$E \cdot 4\pi r^2 = q / \epsilon_0$$

$$E = q / 4\pi\epsilon_0 r^2$$



$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0} \quad (Gauss's Law)$$



23-4 Applying Gauss's Law

3. Spherical Symmetry

3.2 Positive electric charge Q is distributed uniformly throughout the volume of an **insulating sphere** with radius R .

Find the magnitude of the electric field at a point P a distance r from the center of the sphere.

Solution:

$$r > R, \oint \vec{E} \cdot d\vec{A} = Q / \epsilon_0$$

$$E \cdot 4\pi r^2 = Q / \epsilon_0$$

$$E = Q / 4\pi\epsilon_0 r^2$$

$$r < R, \oint \vec{E} \cdot d\vec{A} = q / \epsilon_0$$

$$E \cdot 4\pi r^2 = \frac{r^3}{R^3} Q / \epsilon_0$$

$$E = Qr / 4\pi\epsilon_0 R^3$$

22.22 The magnitude of the electric field of a uniformly charged insulating sphere. Compare this with the field for a conducting sphere (see Fig. 22.18).

